

The Social Cost of Traffic Congestion and Countermeasures in Beijing

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ABSTRACT

Based on the applying of congestion cost assessment methodologies, congestion cost calculation process was analyzed and construction of calculation models was deduced. According to the database of city traffic running status and Statistic Yearbook 2010 in Beijing, the social cost of Traffic Congestion was calculated. The result shows that Traffic Congestion costs about 58 billion Yuan RMB (4.22% of GDP) in Beijing in 2010, including time delay cost, extra oil combustion, traffic accident direct economic loss, vehicle loss cost, environment pollutants. The proportion of GDP is higher than the same scale cities, such as New York, London and St. Paul. The most congestion cost comes from trips by car and the time delay cost is the mainly and directly social cost. For saving of vehicle congestion, this paper suggests that government of Beijing should optimize city function construction around the urban railway station to scatter the function in central metropolitan area, and provide more hospital, school and job in nearby area of new railway station to reduce long distance trip demand.

INTRODUCTION

Traffic congestion is a concept which is difficult to quantify, people's awareness of traffic congestion is different in different cities. Common method to evaluate traffic congestion is average speed, road saturation, crowded rate or queue length, etc. United States, Japan and other developed countries have made a lot of researches about how to evaluate traffic congestion degree, for example, United States has established a relatively perfect congestion evaluation system that applies the expedite research report, traffic congestion evaluation system.

Traffic management evaluation system of city (2004) of the Ministry of public security of China, in which defines severe congestion as Green lights have displayed three times but none vehicle crossing the intersection within signal or length of queue that vehicle blocking at the outside lane of the intersection exceeding 400 meters at the intersections without signal. Beijing use the traffic congestion index to measure the level, congestion index is a conceptual value which is the comprehensive reflection of the state of road network or congestion, just as the digital equivalent of congestion (See table 1).

Table 1. Traffic Congestion Index Table of Beijing

Traffic index	Congestion level	Corresponding traffic	Travelling time
0~2	Fluent	Free traffic fluid	Running with the speed limit
2~4	Almost fluent	Congestion in some road	0.2~0.5 times longer
4~6	Light degree congestion	Congestion in part of the ring road or main road	0.5~0.8times longer
6~8	Middle degree congestion	Congestion in large part of ring road and main road	0.8~1.1times longer
8~10	Heavy degree congestion	Congestion in anywhere	1.1~ times longer

MODEL SPECIFICATION

Time Cost

The time cost includes time loss of drivers, passengers and pedestrian, and the economic loss as a result of the freight delay caused by congestion.

Time Cost of Travel

Traffic congestion causes slowdown of vehicle speed, resulting in the extension of travel time, according to the average speed, average delayed time of each travel can be deduced as following formula.

$$T_d = L \times \left(\frac{1}{V_c} - \frac{1}{V_0} \right)$$

Where T_d is average delayed time each travel; L is average travel distance each time per person; V_c is average speed when traffic is congestion; V_0 is average speed when traffic is free.

Value of unit time is calculated according to the average wage and working time.

$$R_h = \frac{I_a}{D_y \times 8}$$

Where R_h is value per unit time; I_a is average wage per year; D_y is the number of working days per year.

Considering the differences of the unit time value in different travel way, therefore, unite time value is calculated by travel mode and corresponding total travel, as following.

$$C_{pt} = \sum_{i=1}^n R_{i,h} \times T_d \times P_{i,c}$$

Where C_{pt} is time cost of passenger; i is the kind of transportation; $R_{i,h}$ is value of unit time of the i transportation; $P_{i,c}$ is total travel of the i transportation.

Freight Delay Cost

The total congestion cost C_t is calculated by transportation expenditure and the proportion of total congestion time in 24 hours every day.

$$C_t = C_{tr} \times \frac{T_d}{24} = C_{tr} \times \frac{L}{24} \times \left(\frac{1}{V_c} - \frac{1}{V_0} \right)$$

It's very difficult to separate the extra congestion cost C_{con} and total congestion cost, so percentage that extra congestion cost takes of total congestion cost as that P_t between extra congestion time and total congestion time was applied to evaluate the C_{con} .

$$P_t = \frac{T_d}{T_t} = \frac{L \times \left(\frac{1}{V_c} - \frac{1}{V_0} \right)}{\frac{L}{V_c}} = 1 - \frac{V_c}{V_0}$$

$$C_{con} = C_t \times P_t = C_t \times \left(1 - \frac{V_c}{V_0} \right)$$

Energy Cost

Under the condition of congestion, the vehicle run more fuel for longer driving time, at the same time, the vehicle keep braking, starting, whistling, the action as well as low level, low speed when the engine and vehicle matching is not in the ideal condition and other reasons, will increase fuel consumption. Figure 1 is a curve show the relationship between average speed and fuel consumption, It shows clearly that fuel consumption in low or idling speed (20km / h) is almost 1.5 times more than that in normal driving (city normal running speed is 40 ~ 60 km / h)

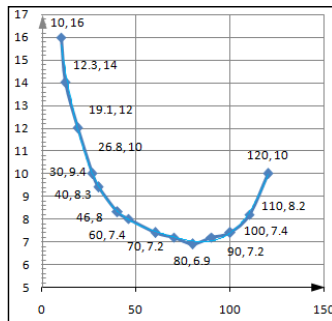


Figure 1. Relationship between average speed and fuel consumption

Fuel consumption in congestion minus the consumption in normal condition is the additional energy consumption, multiply by unit price of energy consumption, the formula shows as following.

$$C_{ex,g} = \sum_{i=1}^n \frac{L_{i,to}}{100} \times (V_{i,con,g} - V_{i,0,g}) \times Y_i$$

Where I is indicates different vehicle types ($i = 1, 2, \dots n$); $L_{i,to}$ is total congestion distance corresponding to the vehicle type; $V_{i,con,g}$ is fuel consumption of different type of vehicle in a jam; $V_{i,0,g}$ is fuel consumption in 100 miles of different type of vehicle; Y_i is unit price of different fuel consumption.

Traffic Accident Cost

Traffic congestion increases more crashes such as scraping and tracing. The accident cost insurance company determines the total cost of traffic accident. Considering traffic congestion increase probability of accident and the insurance company can't claim all the accident loss, the factor ξ_a is multiplied to correct the final results, following is the formula.

$$C_{ac} = C_{to,a} \times P_t \times \xi_a = C_{to,a} \times \xi_a \times \left(1 - \frac{V_c}{V_0}\right)$$

Where $C_{to,a}$ is the accident cost caused by congestion, ξ_a is the accident cost insurance company claims, C_{ac} is the correction coefficient.

Vehicles Loss Cost

Driving in the congestion conditions will increase the fuel consumption, the vehicle use time, and accelerates the wearing of braking disc and tire, speed up the frequency of vehicle maintenance. Congestion maintenance costs can be calculated by the proportion congestion time accounted for travel time. The additional loss cost due to congestion is expressed as:

$$C_d = C_0 \times A_t \times P_t = C_0 \times A_t \times \left(1 - \frac{V_c}{V_0}\right)$$

Where C_0 is vehicle average maintenance costs for congestion, the total number of vehicle is A_t .

Environmental Pollution Cost

Vehicle in congestion keep starting, braking and the condition that low level, low speed increases fuel consumption and exhaust emissions. At low speeds, the work situation of the engine do not reach economic region, leading to the fuel not burning adequately. Figure 2 is the graph of engine external characteristic; it shows that the optimal fuel economy is not in the low speed region, but at the n_3 point.

Driving at low speed or idle pollutes more, when the speed is under 90 km/h, CO, HC of light vehicle increases as the speed increases, so pollution coefficient ξ_e is multiplied to modify the final results. Calculation formula is expressed as:

$$C_e = C_{envi} \times P_t \times \xi_e = C_{envi} \times \xi_e \times \left(1 - \frac{V_c}{V_0}\right)$$

Total Congestion Cost

$$C_{con} = C_{pt} + C_{con} + C_{ex.g} + C_{ac} + C_d + C_e$$

ESTIMATION METHODS AND PROCEDURES

Beijing city was applied to testify the calculation model, conducting the calculation of traffic congestion cost, the basic parameters (including average distance of Beijing citizens’ commuter, urban average speed in the morning and evening peak) involved in the calculation are in Table 2, value of unit time and travel times are shown in Table 3 and Table 4.

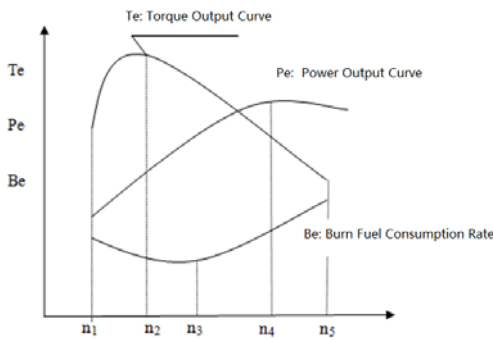


Figure 2. Fuel consumption rate under the different working condition

Table 2. Basic Parameters of Time Cost

Item	Parameter	Number	Unit
Average distance of Beijing citizens’ commuter	L	19.3	Km
Urban average speed in the morning and evening peak	V_c	23.85	Km/h
Urban average speed when traffic is free	V_0	40.0	Km/h
Average income per year	I_a	50415	Yuan
Number of working days per year.	D_y	243	Day

Table 3. Unit Time Value in Different Way of Travel

Item	Parameter	Number	Unit
By public traffic	$R_{1,h}$	26	Yuan/h
By taxi	$R_{2,h}$	31	Yuan/h
By car	$R_{3,h}$	39	Yuan/h

Table 4. Travel Times in Different Way of Travel

Item	Parameter	Number	Unit
By public traffic	$R_{1,c}$	756.2	Million
By taxi	$R_{2,c}$	138.0	Million
By car	$R_{3,c}$	660.9	Million

Calculation of Time Cost

Passenger's Time Cost

Substituting $R_{1,h}=26$, $R_{2,h}=31$, $R_{3,h}=39$, $T_d=0.911$, $P_{1,c}$ = RMB 756.216 million, $P_{2,c}$ = RMB 138.024 million, $P_{3,c}$ = RMB 660.960 million into formula, the passenger's time cost was calculated as following formula, and detail results are showed in Table 5.

$$C_{pt} = \sum_{i=1}^3 R_{i,h} \times T_d \times P_{1,c} = 452.93 \text{ billion}$$

Table 5. Congestion Cost Statistics in Different Way of Travel

	Bus	Taxi	Car	Total amount
Cost in different way of travel (billion)	179.1	38.9	234.8	452.9
Proportion (%)	39.5	8.6	51.8	100.0

Freight Delay Cost

GDP in Beijing in 2010 is 1.37779 trillion Yuan. Logistics cost takes 17.8% of GDP, transportation cost takes 53.1% of the logistics cost, therefore, freight delay cost was calculated as following.

$$C_{con} = C_t \times P_t = C_t \times \left(1 - \frac{V_c}{V_0}\right) = 20.0 \text{ billion Yuan}$$

Energy Cost

In 2009, energy consumption is 65.777 million tons standard coal in Beijing. According to the growth rate 3.96% in 2009 we can estimate that the total energy consumption in 2010 is 68.382 million tons of standard coal. So the transportation energy consumption will reach 13.6767 million tons. Table 6 shows the energy

consumption for traffic congestion differs with the Vehicle model.

Table 6. Consumption of different vehicle

Item	<i>Parameter</i>	Unit	Bus	Taxi	Car
Total traffic mileage	$L_{i.to}$	Million miles	502.3	151.3	33447.1
Congestion consumption	$V_{i.con.g}$	L/km	69.2	9.7	10.2
Fluent consumption	$V_{i.o.g}$	L/km	51.56	7.5	7.6
Price	Y_i	Yuan/Liter	7.15	7.3	7.3

$$C_{ex.g} = \sum_{i=1}^3 \frac{L_{i.to}}{100} \times (V_{i.con.g} - V_{i.o.g}) \times Y_i = 70.1 \text{ billion}$$

Traffic Accident Cost

In 2010 the insurance company claims for traffic accident compensation reaches about 1.8 billion Yuan. The factor ξ_a takes 1.05, so we can calculate the cost for traffic accident.

$$C_{ac} = C_{to.a} \times P_t \times \xi_a = 7.8 \text{ billion}$$

Vehicles Loss Cost

According to the data, average maintenance cost is 1402 Yuan per vehicle, 1/3 time is driving in the state of congestion, so the average congestion maintenance cost is estimated 467.3 Yuan. In Beijing, the total number of annual average vehicle is 4.5 million in 2010. The additional loss cost produced by the congestion was calculated as following.

$$C_d = C_0 \times A_t \times P_t = 8.5 \text{ billion}$$

Environmental Pollution Cost

Main proportion of the money spends on treatment of car emissions. According to the statistics of Beijing environment protection, approximately 73% HC (including volatile organic compounds VOC), 50% NOx and 63% CO come from vehicle exhaust emissions, so it is 40% atmospheric pollution in Beijing. According to estimation of private cars in Beijing, congestion attributes to CO2 emissions increasing by 2.08 million tons all year.

The statistics show that environmental management cost was 33 billion Yuan in 2010 in Beijing. Based on 40% of the traffic pollution, traffic environmental cost will be 13.2 billion Yuan in 2010. At the condition that 1/3 time is driving in the state of congestion, total pollution cost is 4.4 billion Yuan. The factor ξ_a takes 1.15; extra environment cost for congestion was calculated as following.

$$C_E = C_{envi} \times P_t \times \xi_e = 20.4 \text{ billion}$$

RESULTS

According to the calculated data we can get the comparison table with all kinds of traffic cost (see Table 7).

Table 7. Comparison Table of Congestion Cost in Beijing

Congestion cost	Direct costs					Indirect costs	Total
	Time cost		Energy cost	Acci dent cost	Vehicle loss cost	Environmental pollution cost	
	Passenger	Freight					
Number(bil lion Yuan)	452.9	20.0	70.1	7.8	8.5	20.4	579.8
Proportion of GDP(%)	3.29	0.15	0.51	0.06	0.06	0.15	4.22

CONCLUSTION

The calculation model specification of economic congestion costs was presented, and calculated the congestion cost of Beijing city in 2010, from which we can get the following enlightenment:

(1) According to the calculation model that presented in this paper the traffic congestion cost is 57.98 billion Yuan, which is slightly lower than (FENG Xiaoshan et al., 2011), which congestion cost is 72.5 billion Yuan, higher than the 14.6 billion Yuan in the literature. Considering the congestion getting heavier from 2004 to 2010, the higher congestion cost than that literature 7 is reasonable, and (FENG Xiaoshan et al., 2011) takes per capita congestion time while this paper adopts average speed at flat peak to calculate. On contrast, average speed at flat peak can be investigated and accessed more easily through the testing equipment has been installed.

(2) Compared with similar city around the world, the proportion traffic congestion cost accounts for the GDP, the proportion in Beijing (4.22%) is far higher than other similar cities, New York is 0.49%, St. Paulo is 0.62%. It means the traffic congestion in Beijing is rather serious.

(3) From table 6, the car congestion cost accounts for 51.8%, higher than the 35% that the percentage people who use car, indicates that car congestion cost is much higher than the average level of various transportation, unit time value of driving private car is 50% higher than the public traffic affects the results and calculation process. Compare to the people by public transportation, the numbers of travellers who have relatively high income increase in Beijing, traffic congestion cost ratio of car users will gradually decline.

(4) Seen from table 7, time cost is the main cost of traffic congestion, it accounts for about 82% of total traffic cost, therefore, the time cost is the most

important and intuitive among the traffic congestion. At the same time, it is worth noting that in addition to traffic congestion economic cost, environmental pollution caused by traffic congestion (especially atmospheric pollution) and the resulting anxiety cost are the same high with other corresponding social cost.

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